

Getting started with Generative AI

Syllabus-Sync

Logo

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# Solution Brief

## AI-based Deadline Extraction and Calendar Integration for Moodle

### Executive Summary

University students often miss deadlines because assignment information is spread across various Moodle files in different formats and languages. While some deadlines are visible in the Moodle calendar, many are only embedded within course documents, such as PDFs or HTML pages, and are therefore not displayed in the calendar. Based on observations of real Moodle course dashboards, this visibility gap affects around half of all relevant academic deadlines.

This project introduces Syllabus-Sync, a Generative AI-based assistant that extracts deadlines and relevant metadata from uploaded course documents and exports them as structured calendar events.

The solution has a privacy-preserving, user-controlled workflow that enables students to import deadlines into their personal calendars without requiring direct access to Moodle or email accounts.

### Business Value

- **Efficiency:** Automated extraction eliminates the need to scan course documents manually.
- **Reliability:** Fewer missed deadlines thanks to structured calendar reminders.
- **Flexibility:** Works across multiple document formats and languages.
- **Visibility gap reduction:** Any deadlines mentioned only in documents and not shown in the Moodle calendar are made visible and actionable.
- **Strategic value:** Improves academic self-management without requiring deep system integration.

**Quantifiable impact:** Reduces the number of missed or late submissions caused by deadlines that are embedded in documents but not shown in the Moodle calendar.

**ROI estimate:** Saving just 20–30 minutes per course, while preventing even one missed deadline per semester, would already outweigh the cost of using the solution.

**Lecturer return on investment:** With explicit consent, extracted and structured deadline data could be forwarded to departmental secretariats. This would reduce

repetitive administrative work, improve consistency across communication channels and increase overall scheduling reliability.

## Use Case Description

Users:

- **Primary:** University students
- **Secondary:** Study coordinators and lectures (conceptual)

Beneficiaries:

- Students (saving time, reliable planning)
- Academic programmes (fewer late submissions)
- Administrative units (potential reduction in follow-up work)

As-is-process:

- Deadlines are currently communicated via various channels, including Moodle assignments, PDF syllabuses, HTML pages, SharePoint folders (e.g. exam schedules) and room reservation systems.
- Students must manually cross-check several platforms to identify relevant dates.
- Many deadlines mentioned only in documents never appear in the Moodle calendar.
- Calendar entries are created manually, which can lead to errors and omissions.

To-Be Process (MVP):

1. The user uploads course files in PDF, Word, HTML or Excel format.
2. Generative AI then extracts:
  - a. Deadlines
  - b. Task type (exam, assignment or presentation).
  - c. Language-specific date formats
3. The user then reviews and confirms the extracted events.
4. The events are then exported as an .ics file.
5. The user then imports the events into a personal calendar application.

Integration:

Calendar integration is achieved via iCal export (.ics), ensuring compatibility with common calendar applications without requiring direct account access.

## Functional scope

### Must-have (MVP):

- Upload course documents (PDF, Word, HTML, Excel)
- Extract deadlines (date, time, task description)
- Multi-language support (German, English, French, Italian and Spanish)
- Date format normalisation (e.g. “12th January 2026”, “12.01.2026”, January 12, 2026)
- Task prioritisation (exam, assignment, presentation)
- iCal (.ics) export and import
- Follow-up reminders
- Time zone handling
- Manual user review before export

### Could-Have (Out of MVP Scope)

- Extraction of course sections, rooms and lecture times
- Chat between students, lecturers and administration
- Automatic calendar synchronisation
- Data selection and filtering options
- Room booking verification
- Holiday and block week detection
- Extended “AI tutor” functionality

### Not-Have (Explicitly Excluded)

- Direct login to Moodle or email accounts
- Persistent data storage (no database)
- Unlimited historical data retention
- Automatic background access to user data
- Full calendar write access without confirmation

Rationale: Clear scope control, strong privacy protection and reduced implementation complexity.

## Feasibility

- **Technical:** GenAI performs well with unstructured, multilingual academic documents and implicit deadline descriptions.

- **Operational:** Manual confirmation by the user significantly reduces the risk of incorrect calendar entries.
- **Legal/Privacy:** No credential access is required, nor is any personal data stored.
- **Alternatives:** Relying solely on the Moodle calendar is insufficient as many deadlines are communicated exclusively through documents or external systems, such as SharePoint or HTML-based room reservation tools. A purely rule-based extraction approach was considered, but rejected due to the presence of heterogeneous formats, multilingual content and implicit date descriptions.

## Data requirements

- **Type:** Semi-structured and unstructured academic content, including PDFs of syllabuses and assignment descriptions, HTML pages and spreadsheets.
- **Source:** User-uploaded course materials from Moodle, SharePoint (e.g. exam schedules) and HTML-based room or timetable systems.
- **Quality:** Data quality varies due to inconsistent formatting, missing time information and multilingual content. Default values and confidence indicators are used where necessary.
- **Ownership:** All data remains user-owned. Documents are processed temporarily and not stored afterwards.

## Architecture Proposal

### High-Level Architecture

- File upload
- Text extraction
- GenAI deadline extraction (JSON schema)
- Validation & editing UI
- Calendar export (iCal)

### Key Design Principles

- Human-in-the-loop
- Stateless processing
- No database
- User-controlled data flow

## Implementation Roadmap

- **Initiation:** Definition of MVP scope, supported file types and event schema.
- **Development:**
  - File upload
  - Deadline extraction
  - Validation interface
  - iCal export functionality.
- **Deployment:**
  - Preparing a demo-ready prototype.
  - Testing with real Moodle documents.
- **Optimisation:**
  - Multi-format support
  - Confidence scoring
  - Reminder configuration

## Success Criteria

- **Technical:** ≥85% accurate deadline extraction
- **User:** Import deadlines in less than 2 minutes per course
- **Reliability:** No calendar imports without explicit user confirmation

## Risks & Concerns

| Risk                          | Metric         | Countermeasure                        |
|-------------------------------|----------------|---------------------------------------|
| Incorrect deadline extraction | Error rate     | Manual confirmation                   |
| Ambiguous date expressions    | Flag count     | AI clarification and confidence score |
| Privacy concerns              | Data retention | Stateless processing                  |

## Open Discussion Points

- Should reminder timing be fixed or user-configurable?
- Is the .ics format sufficient, or is direct calendar API integration required?
- Should exam schedules be handled differently from assignments?

- Should lecturers be able to approve the forwarding of extracted deadlines to departmental secretariats?
- How should conflicts between multiple data sources (Moodle, SharePoint and room systems) be prioritised?

## References

- Google Skills – Generative AI Learning Path
- iCalendar Specification
- Google Antigravity Documentation